

An Auditable Agentic LLM Pipeline for Rubric-Based Grading: Evidence from a Real E-commerce Course Dataset

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Abstract

Rubric-based grading is essential for transparency and consistency but becomes a bottleneck in large classes. We present an auditable agentic grading pipeline implemented with the OpenClaw framework and apply it to four rubric-based assignments from an undergraduate e-commerce introduction course at UIT. The pipeline includes dataset normalization, text extraction with OCR when needed, incremental result writing, and automated auditing to guarantee one-to-one mapping between submissions and exported scores. Two human graders independently score a stratified subset of submissions; ground truth is defined as the mean of the two scores. We compare a single-agent grader (Phase 1) and a multi-agent grader (Phase 2) that decomposes grading into content/structure/language specialists and aggregates via a chairman agent. We report inter-rater agreement (QWK) and system-vs-human metrics (QWK/MAE/Pearson) with bootstrap confidence intervals, and analyze systematic score shifts and stability signals (veto/spread) induced by Phase 2. Results show that multi-agent grading does not automatically improve reliability and can introduce assignment-dependent calibration shifts; we provide practical recommendations for calibrating and aggregating multi-agent rubric grading in real classroom deployments.

Keywords: automated grading; rubric-based assessment; agentic AI; large language models; reliability; quadratic weighted kappa

1 Introduction

1.1 Motivation

Rubric-based assessment supports fairness and consistent grading, yet grading at scale remains time-consuming in large classes. In practice, student submissions are heterogeneous (DOCX/PDF/TXT and scanned artifacts), and instructors require not only scores and short feedback but also auditable exports (no missing/duplicate records) for course administration and research use.

1.2 Gap

Automated essay/response scoring traditionally evaluates agreement against human raters using statistics such as weighted kappa/QWK [1,2] and provides evaluation frameworks for validity and operational use [3]. Recent studies explore large language models (LLMs) for instruction- and rubric-grounded scoring [10], but real classroom deployments often require (i) end-to-end pipelines that can handle heterogeneous file formats and OCR, (ii) strict auditability guarantees, and (iii) reliability-first reporting that contextualizes system performance with human inter-rater agreement [3,4]. Meanwhile, multi-agent decompositions can help in some contexts but may also introduce calibration mismatch and aggregation instability [14,15].

1.3 Contributions

- C1. An auditable grading pipeline (normalization → extraction/OCR → grading → incremental export → auditing) built on OpenClaw [18].
- C2. A research-ready artifact set (derived, anonymized) for four assignments from a real UIT e-commerce course, with validated one-to-one submission-to-score mapping.
- C3. Reliability-first evaluation with human inter-rater QWK and AI-vs-GT (QWK/MAE/Pearson) with bootstrap confidence intervals.
- C4. Evidence-based analysis of multi-agent grading: systematic score shifts, veto/spread stability signals, and practical calibration recommendations.

2 Related Work

2.1 Automated scoring and agreement metrics

AES/SAS research commonly measures system–human agreement using Cohen’s weighted kappa variants for ordinal scales [1] and practical interpretation guidelines [2]. Evaluation frameworks emphasize reporting reliability, validity, and operational constraints [3].

2.2 Benchmarks and modeling approaches for AES

Benchmarks such as ASAP popularized QWK-based evaluation for essay scoring [7]. Neural AES methods remain a strong baseline paradigm [8], and survey work summarizes modeling choices and evaluation practices [5,9].

2.3 LLM-based rubric-grounded grading

Prompt-based LLM scoring can be adapted to new rubrics without task-specific training, yet agreement varies by prompting and calibration. Recent work documents both promise and limitations in reliability and bias [10,11,12].

2.4 Multi-agent LLM grading and aggregation

Multi-agent designs decompose grading into specialists and aggregate outputs. Such aggregation can introduce severity mismatch and instability without calibration [14]. Structured multi-agent scoring pipelines have also been proposed [15].

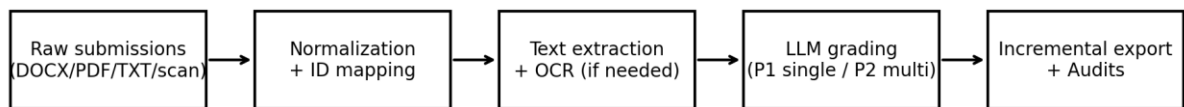
2.5 Reliability-first evaluation

Automated scoring should be interpreted relative to human inter-rater reliability rather than as absolute truth [3].

3 Method

Figure 1 shows the end-to-end pipeline.

Figure 1: OpenClaw grading pipeline (normalization → extraction/OCR → grading → export/audit).



OpenClaw Gateway (gpt-5.2 via Codex OAuth) enforces JSON-only outputs; rounding rules applied post-processing.

3.1 Implementation and model configuration

All LLM calls are served through a local OpenClaw Gateway exposing an OpenAI-compatible Chat Completions endpoint [18]. In our deployment configuration (openclaw.json), the agent's primary model identifier is `openai-codex/gpt-5.2`, accessed via an OAuth-based `openai-codex` profile. This identifier is produced by the gateway routing layer; the underlying model family/weights are proprietary and not publicly disclosed. Therefore, we focus reproducibility on fixed decoding settings, JSON-only scoring schemas, released derived artifacts, and evaluation scripts rather than exact replication of model weights. Phase 1 uses temperature=0.1 and max_tokens=900; Phase 2 uses temperature=0.1 and max_tokens=800 with robust retry logic for rate limits and JSON parsing. Scores are post-processed with deterministic rounding (0.25 per criterion; 0.5 total; range 0–10).

3.2 Phase 1: Single-agent rubric grading

Phase 1 grades all rubric criteria in a single call per submission and outputs per-criterion scores, total score, and short feedback.

3.3 Phase 2: Multi-agent grading (content/structure/language + chairman aggregation)

Phase 2 runs three specialist graders in parallel and aggregates per-criterion scores via a chairman agent (mean per criterion). A veto flag is triggered when specialist totals differ by more than 2.0 points, and a spread value summarizes inter-specialist disagreement.

4 Experimental Setup

We evaluate four assignments (BT1–BT4), each scored on a 0–10 scale with rubric maxima per criterion. Two human graders score a stratified subset (N=30 per assignment) selected to cover performance bands (high/medium/low) based on Phase-1 scores. Ground truth (GT) is defined as the mean of the two human scores. Human annotation budget constraints limited N; we report uncertainty via bootstrap confidence intervals.

4.1 Metrics

We report inter-rater QWK between the two humans and compare Phase 1/Phase 2 against GT using QWK, MAE, and Pearson correlation. For QWK, scores are mapped to integers by multiplying by 2 and rounding, matching the 0.5-point granularity used in totals.

5 Results

5.1 Human agreement and AI-vs-GT (with bootstrap CIs)

Table 1 reports point estimates and 95% bootstrap confidence intervals on the human-scored subset (N=30 per assignment).

BT	Human QWK [CI]	P1 QWK [CI]	P1 MAE [CI]	P1 r [CI]	P2 QWK [CI]	P2 MAE [CI]	P2 r [CI]
BT1	0.698 [0.464, 0.811]	0.484 [0.273, 0.627]	1.108 [0.892, 1.333]	0.808 [0.681, 0.892]	0.043 [-0.026, 0.142]	1.842 [1.517, 2.158]	0.225 [-0.115, 0.557]
BT2	0.684 [0.469, 0.802]	0.331 [0.164, 0.443]	1.375 [1.142, 1.600]	0.694 [0.411, 0.860]	-0.002 [-0.237, 0.244]	0.958 [0.708, 1.250]	-0.010 [-0.356, 0.337]
BT3	0.671 [0.440, 0.835]	0.827 [0.664, 0.904]	0.775 [0.550, 1.042]	0.897 [0.807, 0.960]	0.613 [0.405, 0.747]	1.058 [0.833, 1.300]	0.673 [0.509, 0.837]
BT4	0.628 [0.301, 0.945]	0.720 [0.587, 0.853]	0.575 [0.442, 0.708]	0.753 [0.630, 0.876]	0.517 [0.262, 0.772]	0.608 [0.467, 0.749]	0.553 [0.324, 0.782]

0.791] 0.799] 0.725] 0.861] 0.698] 0.750] 0.725]

5.2 Phase 2 vs Phase 1 score shifts and stability signals (full dataset outputs)

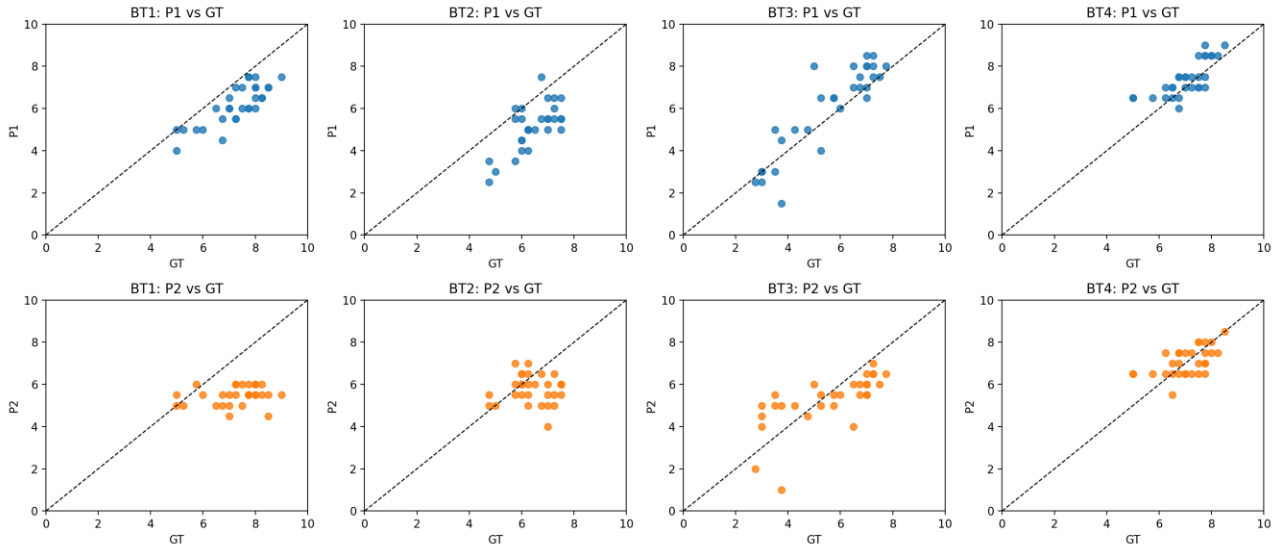
Table 2 summarizes systematic score shifts of Phase 2 relative to Phase 1 on the full exported outputs, along with veto rates and mean spread.

BT	N_full	mean(P2-P1)	median(P2-P1)	veto rate	mean spread
BT1	137	-0.836	-1.000	0.993	5.150
BT2	129	1.787	2.000	0.922	3.387
BT3	123	-0.943	-1.000	0.772	2.653
BT4	131	-0.889	-1.000	0.061	1.133

5.3 Visualization

Figure 2 visualizes agreement with GT for Phase 1 and Phase 2 (N=30 per assignment).

Figure 2: Scatter plots of P1/P2 vs GT across assignments (dashed line indicates $y=x$).



6 Discussion and recommendations

Multi-agent grading does not automatically improve ordinal agreement (QWK) with GT in our setting, and can introduce assignment-dependent score shifts. We attribute this to calibration mismatch between specialists, aggregation smoothing, and surface-feature sensitivity (language) that may be misaligned with rubric intent in factual/sourcing-heavy assignments (BT1-BT2).

Recommendations to improve multi-agent grading reliability:

R1. Calibrate specialist severity using a small anchor set and learn per-grader offsets/scales before aggregation.

R2. Use robust aggregation (median/trimmed mean) rather than simple mean to reduce outlier severity effects.

R3. Rubric-aligned weighting: downweight language/style for BT1–BT2; increase for BT3–BT4.

6.1 Limitations

Our human-scored subset is limited (N=30 per assignment) due to annotator availability; bootstrap CIs can be wide. We did not perform a full sensitivity analysis over temperature and prompting variants; results may change under different decoding settings. Ground truth is defined as the mean of two graders; when graders disagree substantially, adjudication could yield a different reference score.

7 Ethics and privacy

Student submissions are sensitive; we recommend releasing only derived anonymized artifacts (scores/metadata/audit logs) and not releasing raw submissions.

8 Conclusion

We presented an auditable OpenClaw-based grading pipeline and a reliability-first evaluation across four real assignments. Single-agent grading achieved higher agreement with human scores than multi-agent grading in our setting, and Phase 2 exhibited systematic shifts and stability issues. We provided practical calibration and aggregation recommendations for deploying multi-agent rubric grading in real courses.

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